



# CSIT212

# QUANTITATIVE METHODS

## LESSON 2

## Mathematical Modeling

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# Learning Objectives

At the end of this lecture, students should be able to:

- Explain the purpose and components of mathematical models
- .
- Formulate mathematical models from real-world problems.
- Apply mathematical models to solve quantitative problems.
- Interpret and evaluate model results, assumptions, and limitations.
- Use computational tools to analyze and visualize mathematical models.

# Basic Mathematical Model

A **basic mathematical model** is a simplified representation of a real-world situation, system, or problem using mathematical concepts, such as **variables**, **equations**, **functions**, and **relationships**.

It is used to describe, analyze, predict, or solve a problem by translating **real-world information into a mathematical form**.

# Basic Mathematical Model

## General structure (Steps)

- 1) Real-World Problem
- 2) Identify Variables
- 3) Establish Relationships
- 4) Mathematical Model
- 5) Analyze Results
- 6) Interpret Results

# Basic Mathematical Model

## Key Components

| Component           | Description                                         |
|---------------------|-----------------------------------------------------|
| Variables           | Quantities that can change                          |
| Parameters          | Fixed values that characterize the problem          |
| Relationships       | Connections between variables                       |
| Equations/Functions | Mathematical representation of the relationships    |
| Assumptions         | Conditions accepted as true when building the model |
| Output              | Result produced by the model                        |





# Basic Mathematical Model

## Example : PROFIT MODEL

Let  $P$  = Profit (Output/Dependent)

$R$  = Revenue (Input/Independent)

$E$  = Expenses (Input/Independent)

Basic Model.

$$\text{Profit} = \text{Revenue} - \text{Expenses}$$

$$P(R, E) = R - E$$

# Basic Mathematical Model

Let  $X$  = quantity sold (input/independent)  
 $s$  = selling price per unit (constant)  
 $v$  = variable cost per unit (constant)  
 $f$  = fixed cost (constant)

**Revenue** = selling price  $\times$  quantity sold  
 $R(X) = sX$

**Expenses** = fixed costs + variable cost  $\times$  quantity sold  
 $E(X) = f + vX$

# Expanded Mathematical Model

An **expanded mathematical model** is a more detailed representation of a real-world system or problem that incorporates **multiple variables, relationships, constraints, assumptions, and factors** to provide a more realistic analysis or prediction.

Unlike a basic mathematical model, which may describe a simple relationship between variables, an expanded model considers **several interacting factors that can affect the outcome.**



# Expanded Mathematical Model

## EXPANDED PROFIT MODEL

$$P = R - E$$

$$P(R(X), E(X)) = R(X) - E(X)$$

$$P(X) = R(X) - E(X)$$

$$P(X) = sX - (f + vX)$$

$$P(X) = sX - f - vX \quad \text{or} \quad P(X) = X(s - v) - f$$

# Basic and Expanded Mathematical Model

## Example

A company buys, sells, and repairs old clocks. One of its products is rebuilt clock springs, which are sold for \$10 per unit. The company has a fixed equipment cost of \$1,000 to produce the springs. The variable cost of materials for each spring is \$5 per unit. Using the Expanded Profit Model, answer the following:

- (A) Identify the constants in the model.
- (B) Calculate the revenue, expenses and profit when no units are sold.
- (C) Calculate the revenue, expenses and profit after selling 100 units.
- (D) Compute the minimum number of units that must be sold for the company to begin making a profit. What will be the profit?



# Basic and Expanded Mathematical Model

## Answers

(A)  $s = 10$   
 $v = 5$   
 $f = 1000$

(B) At  $X = 0$ ,  
 $R(0) = sX = 10(0) = \$ 0$   
 $E(0) = f + vX = 1000 + 5(0) = \$ 1000$   
 $P(0) = R(0) - E(0) = 0 - 1000 = - \$ 1000$

(C) At  $X = 100$ ,  
 $R(100) = 10(100) = \$ 1000$   
 $E(100) = 1000 + 5(100) = \$ 1500$   
 $P(100) = 1000 - 1500 = - \$ 500$

# Basic and Expanded Mathematical Model

Answers (D)

Model:  $P(X) = X(s - v) - f$

$$f < X(s - v)$$

$$X > \frac{f}{s - v}$$

Goal:  $0 < P(X)$

$$\frac{f}{s - v} < \frac{X(s - v)}{s - v}$$

$$X > \frac{1000}{10 - 5}$$

$$0 < X(s - v) - f$$

$$f < X(s - v)$$

$$\frac{f}{s - v} < X$$

$$X > 200$$

To turn a profit,  $X > 200$ . Hence, we turn a profit at  $X = 201$ .

To check:  $P(201) = (10 - 5)(201) - 1000 = \$ 5$

# Break-Even Analysis

Break-even point (BEP): number of units where **profit = 0**

Formula:

$$BEP = \frac{\text{Fixed Cost}}{(\text{Selling Price} - \text{Variable Cost})}$$

BEP determines the minimum sales needed to avoid losses.

To get minimum profit level:  $X = BEP + 1$  where  $MIN(P(X)) > 0$



# Break-Even Analysis

## Example

Ray Bond sells handcrafted yard decorations at county fairs. The variable cost to make these is \$20 each, and he sells them for \$50. The cost to rent a booth at the fair is \$150.

- (A) How many units must Ray sell to reach the break-even point?
- (B) What is the minimum number of units that must be sold to begin earning a profit?
- (C) What is the minimum profit earned at that sales level?
- (D) What is the total revenue at the minimum-profit level?
- (E) What is the total expense at the minimum-profit level?



# Break-Even Analysis

Answers

$$\begin{aligned}s &= 50 \\ v &= 20 \\ f &= 150\end{aligned}$$

$$(A) \quad BEP = \frac{f}{s-v} = \frac{150}{50-20} = 5$$

$$(B) \quad X = 5 + 1 = 6$$

$$(C) \quad P(6) = (50 - 20)(6) - 150 = \$30$$

$$(D) \quad R(6) = 50(6) = \$300$$

$$(E) \quad E(6) = 150 + 20(6) = \$270$$

# Application

## GROWTH MODEL

A technology company monitors the amount of data stored in its cloud storage system. The amount of stored data increases over time as more users upload files.

The amount of stored data is modelled using an exponential growth function:

$$D(t) = D_0 (1 + r)^t$$

where:

$D(t)$  = amount of data stored after (t) years

$D_0$  = initial amount of data

$r$  = yearly growth rate expressed as a decimal

$t$  = number of years





# Application

(1) Supposed that the growth must be expressed in months ( $m$ ). Write the expanded model.

Hint : Use  $D_0(1 + r_m)^{12} = D_0(1 + r)$

(2) How much data will be stored after 6 months if the startup currently has 800 GB of data stored in its cloud system and data storage increases by 10% every month? What if it increases by 20% annually compounded monthly?

(3) How many months will it take for the company to reach 1,500 GB of stored data assuming 10% rate and initial of 800 GB?

## Application

(1) Supposed that the growth must be expressed in months ( $m$ ). Write the expanded model.

Solution :

$$D_0(1 + r_m)^{12} = D_0(1 + r)$$

$$(1 + r_m)^{12} = 1 + r$$

$$1 + r_m = (1 + r)^{\frac{1}{12}}$$

$$r_m = (1 + r)^{\frac{1}{12}} - 1$$

$$D(m) = D_0(1 + r_m)^m$$

$$D(m) = D_0 \left( 1 + [(1 + r)^{\frac{1}{12}} - 1] \right)^m$$

$$D(m) = D_0 \left( (1 + r)^{\frac{1}{12}} \right)^m$$

$$D(m) = D_0(1 + r)^{\frac{m}{12}}$$

## Application

(2) How much data will be stored after 6 months if the startup currently has 800 GB of data stored in its cloud system and data storage increases by 10% every month? What if it increases by 20% annually compounded monthly?

Solution :

$$D(m) = D_0(1 + r_m)^m$$

$$D(6) = (800)(1 + 0.10)^6$$

$$D(6) = (800)(1.1)^6$$

$$D(6) \approx 1,417.25 \text{ GB}$$

$$D(m) = D_0(1 + r)^{\frac{m}{12}}$$

$$D(6) = (800)(1 + 0.20)^{\frac{6}{12}}$$

$$D(6) = (800)(1.2)^{\frac{1}{2}}$$

$$D(6) \approx 876.36 \text{ GB}$$



## Application

(3) How many months will it take for the company to reach 1,500 GB of stored data assuming 10% rate and initial of 800 GB?

Solution :

$$D(m) = D_0(1 + r_m)^m$$

$$m = \frac{\ln\left(\frac{D(m)}{D_0}\right)}{\ln(1 + r_m)}$$

$$m = \frac{\ln\left(\frac{15}{8}\right)}{\ln(1.1)}$$

$$\ln\left(\frac{D(m)}{D_0}\right) = \ln(1 + r_m)^m$$

$$m = \frac{\ln\left(\frac{1500}{800}\right)}{\ln(1 + 0.10)}$$

$$m \approx 6.6 \text{ months}$$

$$\ln\left(\frac{D(m)}{D_0}\right) = m \ln(1 + r_m)$$

$$m \approx 7 \text{ months}$$



# Application

## Web Traffic Model

A technology company monitors the number of visitors to its website. After launching a new online advertising campaign, the company observes that website traffic increases by approximately 2,500 visitors every month. Before the campaign, the website received 12,000 visitors per month. The company wants to model its website traffic using a linear function.



# Application

- (1) Formulate a mathematical model for the number of website visitors after  $m$  months.
- (2) Under normal conditions, traffic increases by 2,500 visitors each month. The company also conducts advertising campaigns, with each campaign generating approximately 1,000 additional visitors. Each social media promotion generates approximately 750 additional visitors.

Develop an expanded mathematical model for website traffic using:

$m$  = number of months  
 $a$  = number of advertising campaigns  
 $s$  = number of social media promotions.

# Application

(3) How many visitors are expected after 6 months?

(4) If the company's target is 40,000 monthly visitors, how many months will it take to reach the target?

(5) If the company's target is 40,000 monthly visitors, number of campaigns is 3 and number of social media post is 2, how many months will it take to reach the target?



**END OF LESSON**